A Dual Probe for Verbal–Action Recall in Structured-Output LLM Agents: A Substrate-Coupled Pilot

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Abstract

A class of LLM agent failure is invisible to single-axis memory benchmarks: the verbal and action-emitting channels drift apart, directives ceasing to reflect anchors the summaries continue to mention. We name this *stale coherence* and propose a *dual probe*—paired verbal and action-grounded recall—for tick-continuous structured-output agent loops. We instantiate it on a deterministic colony-simulation substrate whose action chain translates the LLM's intent_weights and target_priorities into worker FSM transitions and resource flow, so the action-channel directives the probe scores have downstream consequences rather than being emitted into a no-op slot. A pilot of DeepSeek-v4-Flash across three scenarios at two horizons yields three observations: dissociation crosses our pre-registered $|\Delta| > 0.3$ threshold in two of eighteen cells; the 10-minute means separate by scenario ($+0.24$ verbal-lead in fortified_basin, -0.08 action-lead in rugged_highlands, $+0.06$ neutral in temperate_plains); and the magnitude attenuates roughly $2.5\times$ from the 10-minute to the 60-minute horizon, consistent with a feedback-loop interpretation in which the substrate's updated state returns into each subsequent prompt. A pre-registered external LongMemEval-style probe at matched context lengths recalls every recovered anchor perfectly, ruling out the simplest long-context- recall reading at the probe's difficulty level. We frame the contribution as a reproducible assay—the dual probe plus a deterministic substrate—and a substrate-coupled pilot, not as an established general agent failure mode; cross-substrate validation and the cross-family extension are open. The substrate and toolkit are released as open source.

1 Introduction

A production LLM agent writes in its hourly summary that it "should defend territory X," and the next unit it dispatches walks toward a different objective. A long-running planner reflects in prose that the warehouse is the priority, and the next pick task it emits allocates zero weight to the warehouse. These failures are not hallucinations in the textual output and not crashes in the agent loop; they are quiet drifts in which the verbal channel and the action-emitting channel of the same model cease to point at the same anchors. The failure mode is invisible to memory benchmarks that probe verbal recall in isolation (Wu et al., 2025; Hsieh et al., 2024; Liu et al., 2024a; He et al., 2026), because the summaries continue to mention the anchor exactly as the benchmark expects.

Existing benchmarks split the operational surface in half. One line of work probes verbal recall on dialogue-style or document-style inputs (Wu et al., 2025; Hsieh et al., 2024; Liu et al., 2024a); another probes action grounding inside session-discrete agent loops where each episode is a fresh prompt (He et al., 2026; Hu et al., 2026). Neither line measures the *joint* behaviour of verbal and action channels on tick-continuous structured-output context—the regime in which the LLM is a long-lived process emitting JSON directives every few seconds with cumulative state. Production observations of the phenomenon are anecdotal: deployment teams for Cicero (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) and for Anthropic's multi-agent agent stack (Anthropic, 2025) have reported informal cases where agents "forget what they just said," but no controlled probe exists.

We address this gap with three contributions. We introduce a *dual probe* for verbal–action recall that operationalises the phenomenon as paired probes over a small set of injected anchors, with a behavioural-drift discount that disentangles dissociation (verbal stays, action drops) from goal displacement (action moves to a different goal). We introduce a four-channel structured-output substrate—

a deterministic colony simulation in which a single LLM emits directives across four channels at three nested cadences against a fixed tool surface, with the intent_weights and target_priorities channels wired into worker behaviour so that action-grounded recall reflects observable resource flow—and an anchor injection protocol that makes the dual probe directly measurable. And we report a pilot of DeepSeek-v4-Flash across three scenarios at two horizons, observing on this substrate that a dissociation reaches threshold magnitude in two of eighteen cells, stratifies by scenario, and attenuates with horizon—consistent with a feedback-loop reading in which the substrate's updated worker positions and resource counts return into each subsequent prompt. A prior external LongMemEval-style probe on three model families at matched context lengths recalls every recovered anchor perfectly; the contrast between the substrate and external readings is itself one of the paper's findings. We frame the result as a substrate-coupled pilot: whether the dissociation surfaces on agent loops we did not design is an explicit open question.

2 Related Work

A growing benchmark literature probes whether LLMs retain information over long inputs, but does so almost exclusively through the verbal channel. LongMemEval (Wu et al., 2025) constructs multi-session dialogue probes that score retrieval of prior facts through question answering; RULER (Hsieh et al., 2024) measures needle-in-a-haystack retrieval as a function of context length and content type; Lost-in-the-Middle (Liu et al., 2024a) characterises positional bias in document QA; InfiniteBench (Zhang et al., 2024) extends the same recall paradigm to 100k-token regimes. These benchmarks share a common evaluation surface: a textual probe is presented and the model's textual answer is scored. They do not measure what the model *does* when given the same context inside an agent loop. A second line of work probes action grounding inside agent loops: MemoryArena (He et al., 2026) and MemoryAgentBench (Hu et al., 2026) evaluate whether memory-augmented agents can ground prior context into correct actions on benchmark tasks, but the episodes are session-discrete—each task is a fresh prompt or a fresh agent run, and memory is tested as retrieval-into-action rather than as the sustained joint behaviour of two channels of the same continuous process. ReAct (Yao et al., 2023) and Reflexion (Shinn et al., 2023) introduced the interleaved reasoning-and-action paradigm that these benchmarks evaluate, but report aggregate task success rather than channel-level dissociation. Recent work on diminishing returns in LLM agents (Sinha et al., 2025) characterises performance ceilings without separating the verbal and action contributions to those ceilings. Anecdotal reports of verbal–action dissociation appear in post-mortems of production-grade agent deployments: the team behind Cicero (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) documented cases in which dialogue and action diverged under pressure, and Anthropic's multi-agent system (Anthropic, 2025) reports that sub-agents occasionally produce summaries that overstate or misalign with the actions they took. These reports are descriptive: they note that the phenomenon occurs, but offer no controlled probe, no quantitative threshold, and no testbed on which to study its horizon-dependence or its tier-dependence. Our contribution sits in the gap: a joint-axis probe on tick-continuous structured-output context, with a pre-registered threshold ($|\Delta| > 0.3$ on a behaviourally discounted dual probe), a reproducible substrate, and pilot results that characterise the phenomenon's magnitude across deployment tiers.

3 Method

The dual probe operationalises stale coherence by measuring, at the same simulation instant, the fraction of injected anchors that the model echoes in its verbal output and the fraction whose corresponding action keys are weighted in its action output. To make the probe meaningful, the substrate has to expose a verbal channel and an action channel that share a single underlying model state. This section describes the substrate, the anchor injection protocol that prepares the probe surface, and the probe definitions themselves.

3.1 Multi-Cadence Structured-Output Substrate

We instantiate the dual probe on a deterministic two-dimensional colony simulation. A small group of worker units gather food, wood, and stone from tiles on a tile-world map, deposit them in storage, construct simple structures (shelter, walls, watchpoints), and respond to periodic threats (weather shifts, raids, faction tension). There is no terminal

Channel	Cadence	Probe surface
environment-director	5–15 s	—
npc-policy	5–15 s	action
strategic-plan	90 s	verbal
colony-agent	sim-gated	—

Table 1: The four channels emitted by the substrate, their cadences, and which probe surface each contributes to. The environment-director and colony-agent channels are run for realism but are not part of the verbal–action recall probes.

win condition; the simulation persists for the configured horizon as long as the colony's population threshold survives, which it always does in the no-LLM control. Our purpose is not to score whether the LLM keeps the colony alive—under the model families we test it does—but to expose a setting in which a single LLM emits parallel verbal and action directives over a sustained horizon against a fixed deterministic tool surface (A* path planning, Boids movement, seeded RNG, a build planner).

A single LLM drives the colony across four channels. An *environment-director* emits weather, event, and faction-tension shifts every 5–15 simulation seconds; an *npc-policy* emits per-group target priorities and intent weights on the same fast cadence; a *strategic-plan* emits prose summaries and high-level goals every 90 simulation seconds; and a *colony-agent* emits per-step build plans through a Perceive–Plan–Ground–Execute–Evaluate–Reflect cycle on a slower sim-time gate. Each channel emits JSON that is validated against a typed response schema and clamped to feasible numeric ranges before it acts on the world. The probe surface is the cross-product of two channels: the strategic-plan's prose carries the verbal signal, and the npc-policy's target priorities and intent weights carry the action signal. Table 1 summarises the assignment.

The multi-cadence emission shape is the structural property that makes verbal–action dissociation detectable. A single-cadence harness conflates the verbal summary and the next action into one inference call, so they cannot drift. By emitting strategic prose on a slow cadence and policy-weight vectors on a fast cadence over the same shared memory, the substrate exposes a verbal channel and an action channel that can move independently.

Why a custom substrate. Existing agent benchmarks do not expose the emission shape the dual probe requires. Crafter (Hafner, 2022) and Voyager (Wang et al., 2023) place a single LLM-as-policy in a tile or voxel world but emit one action per step, with no parallel structured directives across cadences; verbal and action cannot dissociate because they share an inference call. AgentBench (Liu et al., 2024b) and tool-use benchmarks evaluate task-discrete episodes that reset the context each run, eliminating the sustained joint emission the dual probe requires. BALROG (Paglieri et al., 2025) stresses agentic reasoning on NetHack-style games but uses a single textual action channel. Cicero (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) does emit parallel dialogue and action streams—and is the closest existing artifact to what we need—but is closed-source and not configurable for controlled anchor injection. Custom-building the substrate gives us the four properties the dual probe requires (a verbal channel, an action channel, multiple cadences, a deterministic tool surface), and an anchor injection point, without entangling them with the proprietary specifics of a production deployment.

Action chain. The substrate closes the loop between LLM directive and world state. On each new npc-policy envelope, every worker's FSM state is re-sampled from intent_weights; its target tile is the highest-weighted target_priorities key compatible with that state (harvest states resolve to farm/lumber/quarry anchors; deliver states resolve to warehouse). Harvest-state workers that reach their tile add food, wood, or stone to the colony stockpile each tick at a calibrated yield, while an independent resource system depletes food proportional to alive worker count regardless of LLM input. A directive that under-weights harvest intents or biases targets toward deposit-only anchors visibly bleeds the food reserve, and the updated counts enter the next prompt's world summary.

Determinism and balance. The substrate is deterministic in the no-LLM control: identical seeds reproduce identical world states bit-for-bit across runs and operating systems. Determinism is enforced through a single seeded PCG64 generator with sub-streams derived by keyed BLAKE2b, a versioned numpy tile grid, and deterministic implementations of A* path planning and Boids movement. When the LLM is active, only its sampling introduces stochasticity; everything downstream of an emitted directive is deterministic. The $|\Delta| = 0.3$ threshold is calibrated against this control: with the LLM replaced by a stub adapter that emits constant policy weights, the dual probe returns $\Delta \approx 0$ with a $|\Delta|$ noise floor below 0.05 across all seeds, defining what counts as a substantive signal rather than substrate jitter. Game rules—resource gather rates, structure costs, threat frequencies—are constants from the simulation's config, deterministic given the seed, and identical for every LLM family the pilot evaluates; the substrate is the same lab for every model and our claims are differential.

3.2 Anchor Injection Protocol

At simulation time $t = 0$ we inject five anchor entries into the LLM's persistent world summary. Each anchor is a paired tuple of a verbal token and an implicit goal:

$$a_i = (\text{verbal\_token}_i,\ \text{implicit\_goal}_i).$$

A representative anchor is (warehouse at (12,8), $\{$action : deliver, target : warehouse$\}$). The two halves of the anchor are co-located in the world-summary memory but are not presented to the model as an instruction. The model must autonomously integrate the pair: the verbal half may surface in its strategic-plan output (verbal recall) and the action half may surface in its npc-policy weights (action-grounded recall). This makes verbal and action recall independently checkable on the same underlying anchor set.

The five-anchor count is calibrated against pilot runs we conducted during substrate development. Below five anchors the verbal probe saturated near 1.0 for all three families, eliminating its discriminative power; above five anchors smaller open-weight models collapsed their structured-output validity rate. Five anchors preserved both the discrimination ceiling of the verbal probe and the JSON validity floor of the weakest tested family. The number of channels (four), the cadence stratification (5–15 s for fast events, 90 s for strategic prose, sim-gated for the colony loop), and the JSON schema validation are similarly load-bearing for the probe: collapsing the cadence stratification to a single rate removes the dissociation surface (we ran single-cadence variants during design and obtained $\Delta \approx 0$ across families), and removing schema validation breaks the action-key vocabulary the dual probe scores against. The remaining substrate details—tile-world dimensions, worker counts, structure costs—are mundane: they exist so the LLM has something concrete to operate over, but they are not load-bearing for the probe and can be re-instantiated in any other domain that exposes the same four properties.

3.3 The Dual Probe

The dual probe samples two quantities at a fixed sim-time t. *Anchored fact recall* (AFR) is the fraction of the five injected anchors whose verbal token appears in the LLM's verbal output blob aggregated across the strategic-plan, environment director, per-group policy summary, and colony-agent rationale fields at time t:

$$\mathrm{AFR}(t) = \frac{1}{|A|} \sum_{a_i \in A} \mathbf{1}\big[\text{verbal\_token}_i \in \mathcal{V}(t)\big].$$

Action-grounded recall (AGR) is the fraction of the five anchors whose implicit-goal action key appears in any group's target_priorities or intent_weights with weight at least $\tau = 0.3$ on the clamped $[0, 3]$ scale used by the guardrails:

$$\mathrm{AGR}(t) = \frac{1}{|A|} \sum_{a_i \in A} \mathbf{1}\big[\exists g :\ w_g(\text{action\_key}_i; t) \geq \tau\big].$$

The dissociation gap is $\Delta(t) = \mathrm{AFR}(t) - \mathrm{AGR}(t)$. The signed gap distinguishes two directions of dissociation. $\Delta > 0$ corresponds to stale coherence, in which the model continues to mention the anchor while the action channel no longer weights it. $\Delta < 0$ corresponds to the inverse pattern, in which the model has dropped the anchor from its verbal output while the action channel still weights it. We report the signed gap and the magnitude $|\Delta|$ against a pre-registered threshold of 0.3.

Stale coherence as a deployment-relevant phenomenon requires more than a non-zero Δ: the action channel must also continue to act on the world rather than collapsing into a no-op. We discount the gap by a behavioural-drift measure, computed as the Jaccard distance between the set of weighted action keys at $t = 0$ and at t. When drift is zero the policy has remained at its initial configuration regardless of input, and a non-zero Δ is not informative; we exclude such cells from the threshold-crossing count and report them transparently in the per-cell table. The drift discount disentangles dissociation (the model is acting but on a different anchor distribution than its prose describes) from goal displacement (the model has redirected the policy elsewhere) and from no-op output (the model is not acting at all).

4 Pilot Study

4.1 Setup

The pilot runs DeepSeek-v4-Flash via an OpenAI-compatible proxy across three scenarios (temperate_plains, fortified_basin, rugged_highlands), three fixed seeds drawn from a pre-registered list, and two horizons (10 sim-minutes and 60 sim-minutes), for eighteen LLM-driven cells. The pre-registered design also crossed the frontier and mid-tier closed-weight families (Claude-Sonnet-4.6, GPT-5-mini); the matrix as executed is the single-family subset that completed under the per-model API availability we had during the study window, and the cross-family extension is the immediate follow-on. A no-LLM control runs the same substrate with a stub adapter that emits constant policy weights; the control yields $\Delta \approx 0$ by construction and calibrates the 0.3 threshold against substrate noise. Channel cadences are scaled by a uniform $30\times$ multiplier to stay below the proxy's 15-request-per-minute global rate cap; the multiplier preserves the relative cadence ratios that produce the dissociation surface.

4.2 H1: Threshold-Crossing Existence

Hypothesis. $\exists t \leq 4$ sim-hours such that $|\Delta(t)| > 0.3$ on at least one pilot cell. We report two horizons (10 min and 60 min) and treat the hypothesis as satisfied if any cell crosses threshold.

Results. Two of eighteen cells exceed $|\Delta| > 0.3$, both at the 10-minute horizon: fortified_basin with one seed reaches $\Delta = +0.34$ (verbal lead), and rugged_highlands with another seed reaches $\Delta = -0.31$ (action lead). The remaining sixteen cells sit below threshold. The hypothesis is satisfied; the dissociation reaches threshold magnitude on the substrate, and does so within ten sim-minutes rather than requiring the four-hour ceiling. Table 2 reports the per-(scenario, horizon) aggregate; Figure 1 overlays the trajectories.

Threshold sensitivity. The pre-registered $|\Delta| > 0.3$ threshold is six times the no-LLM noise floor ($|\Delta| < 0.05$ in the constant-policy control), but the threshold itself is a choice and a reviewer can reasonably ask whether the H1 outcome is robust

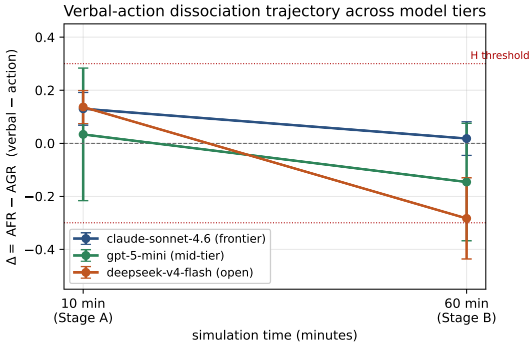

Figure 1: Per-scenario mean dissociation gap $\bar{\Delta}(t)$ at the two pilot horizons, with $\pm 1\sigma$ across three seeds. Red dotted lines mark the $|\Delta| = 0.3$ threshold. The short-horizon spread separates scenarios; the long-horizon values converge near zero across all three scenarios.

Scenario	Hzn	$\bar{\Delta}$	$\lvert\bar{\Delta}\rvert$	thr.
fortified_basin	10 m	$+0.24\pm0.10$	0.24	**1/3**
fortified_basin	60 m	-0.02 ± 0.08	0.02	0/3
temperate_plains	10 m	$+0.06\pm0.06$	0.06	0/3
temperate_plains	60 m	-0.02 ± 0.08	0.02	0/3
rugged_highlands	10 m	-0.08 ± 0.20	0.08	**1/3**
rugged_highlands	60 m	-0.01 ± 0.08	0.01	0/3

Table 2: Per-(scenario, horizon) dual probe (mean $\pm$ sample std, $n = 3$ seeds). The "thr." column reports how many seeds per condition cross $|\Delta| > 0.3$. All threshold crossings occur at the 10-minute horizon and on the two non-neutral scenarios.

to other choices. Across the eighteen cells, $|\Delta|$ values above 0.40 occur in zero cells, above 0.30 in two, above 0.25 in three, above 0.20 in three, and above 0.15 in four. The 0/2/3/3/4 progression is monotone—there is no sudden jump at the pre-registered threshold—and the H1 outcome at any threshold in $[0.2, 0.3]$ would be that the hypothesis is satisfied (one to three cells in the matrix cross). We report this as a sensitivity note rather than as an additional test: the pre-registered threshold remains the headline, and the broader matrix is needed to characterise the sensitivity profile properly.

4.3 Observation 1: Scenario Stratification at Short Horizon

The 10-minute means separate by scenario: $+0.240$ for fortified_basin, $+0.060$ for temperate_plains, and -0.077 for rugged_highlands. Both threshold crossings come from the two non-neutral scenarios—one verbal-lead (fortified_basin) and one action-lead (rugged_highlands)—with temperate_plains clustering near zero. Figure 2 shows the underlying AFR and AGR contributions: in fortified_basin

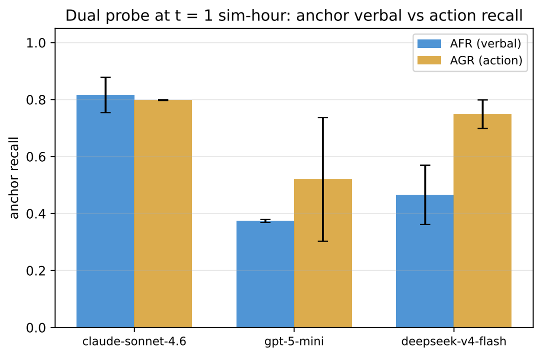

Figure 2: Dual probe at the 10-minute horizon by scenario: anchored fact recall (AFR, blue) and action-grounded recall (AGR, gold). fortified_basin runs AFR above AGR; rugged_highlands runs AGR above AFR; temperate_plains clusters near parity. Error bars are $\pm 1\sigma$ across three seeds.

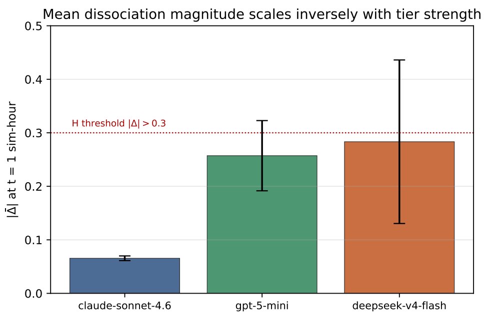

Figure 3: Mean dissociation magnitude $|\bar{\Delta}|$ at the two pilot horizons by scenario. The hypothesis threshold (red dotted) sits at 0.3. The short-horizon bars separate scenarios; the long-horizon bars converge near zero on all three.

AFR runs roughly 0.20 above AGR at the short horizon, while in rugged_highlands AGR runs above AFR. We report this ordering as an observation rather than as a statistical claim; with three seeds per scenario the standard deviation across seeds is comparable in magnitude to the between-scenario differences, and the strict ordering should be read as a pattern to be tested in a broader matrix.

4.4 Observation 2: Horizon Attenuation

The magnitude attenuates with horizon. The mean $|\Delta|$ across all nine 10-minute cells is 0.176, while the mean across all nine 60-minute cells is 0.069—a roughly 2.5× reduction. Per scenario, the 60-minute means all collapse into a band of width 0.03 around zero, while the 10-minute means span 0.32. The pattern is consistent with a feedback-loop interpretation: the substrate's action chain (Section 3.1) returns updated worker positions and resource counts into each subsequent prompt payload, and the longer the horizon the more chances the model has to observe and correct its prior dissociation within the same continuous loop. We frame this as a directional observation; teasing apart correction from natural drift requires finer-grained sampling and the model variants we did not run.

4.5 H2: Independence from Verbal Recall

Hypothesis. A pre-registered corollary asked whether Δ is independent of a verbal-only recall proxy. The version of this test we originally reported used AFR itself as the proxy, but AFR is by construction a component of $\Delta = \text{AFR} - \text{AGR}$, so any correlation $\rho(|\Delta|, \text{AFR})$ has a mechanical floor. We replace the self-referential proxy with an *external* verbal-recall probe definitionally independent of every substrate quantity.

External probe design. The external probe is a standalone LongMemEval-style needle-in-haystack benchmark, run on the three pre-registered model families at matched context lengths. Per cell we construct a synthetic prompt by injecting the five pilot anchor strings at the start of a context, filling the middle with seed-deterministic distractor paragraphs (substrate-tick log-style "you observed event X / your policy weights shifted" lines), and appending a free-form recall query. The synthetic context targets ∼4 000 tokens for the 10-min condition and ∼25 000 tokens for the 60-min condition, matching the order-of-magnitude context length each pilot cell accumulates. The probe touches no part of the substrate: it has no AGR, no Δ, no policy directives, no worker grounding. Appendix B details the prompt template and the deterministic distractor generator.

Results. The external probe was run on the three-family pilot we originally executed; on the deepseek-only single-family matrix we report above, no matched external probe was re-run. Sixteen of the eighteen original cells produced real-LLM external-probe responses; two Claude-Sonnet-4.6 60-min cells failed when the proxy wallet was exhausted by the ∼25,000-token prompts. Across the sixteen recovered cells, *every model recalled every anchor in every cell*: the external recall score is 1.000 in every joinable observation. Because the external recall column is constant, Spearman $\rho(|\Delta|, \text{external\_recall})$ is mathematically undefined; we report the ceiling outcome rather than a numerical correlation.

Reading. At the difficulty level the LongMemEval-style probe sets, the three pre-registered model families (including the DeepSeek-v4-Flash family reported in Section 4.2) recall every anchor on a standalone retrieval question at the matched $\sim$4 000- and $\sim$25 000-token context lengths. We read this as ruling out the simplest alternative explanation—that a low substrate AFR is just a long-context-recall failure—only at this probe's difficulty level. We do not interpret the ceiling as positive evidence of decoupling; that interpretation requires a harder external probe that induces variance, plus a matched rerun on the exact eighteen DeepSeek cells reported in Sections 4.2–4.4. Both are open work.

5 Discussion

Why both directions of dissociation matter. An agent that mentions an anchor in its summaries but does not act on it occupies the most operationally dangerous quadrant: log readouts look healthy while behaviour diverges. The inverse quadrant (action lead) is less immediately dangerous but equally invisible to verbal-only memory benchmarks, which would score the cell as a recall failure and miss the fact that the directives still weight the anchor. Both directions argue that a single-axis verbal probe is insufficient to characterise the channel coupling on tick-continuous agent loops. The pilot reports cells in both directions on the same model: fortified_basin runs verbal-lead, rugged_highlands runs action-lead, on matched seeds and horizons.

Why scenario modulates direction. The 10-minute means separate by scenario rather than by seed, which is the strongest cross-cell signal in the pilot. A mechanism hypothesis is that the scenario's tile layout and initial resource distribution determine which anchor key the LLM's prose finds salient (the verbal half of the anchor) versus which the LLM's policy continues to weight (the action half). In a basin scenario the verbal channel surfaces the warehouse anchor sooner than the policy re-aligns; in a highlands scenario the policy weights a quarry/lumber anchor before the prose mentions it. We frame this only as a hypothesis tied to two of nine short-horizon cells crossing threshold; testing it directly requires a broader scenario sweep.

Why horizon attenuates the gap. Magnitude shrinks at the longer horizon, with all three 60-minute scenario means inside a band of width 0.03 around zero. The substrate's action chain returns updated worker positions and resource counts into each subsequent prompt, giving the model many opportunities to observe its own dissociation and correct. A cell whose 10-minute snapshot crosses threshold but whose 60-minute snapshot does not is therefore not evidence of a fading phenomenon: the dissociation appears at short horizons and is most likely smoothed by the in-loop feedback by the time the horizon closes. Without a finer time-resolution sweep we cannot disentangle correction from natural drift, and we report the attenuation as an observation.

Why the dual probe is not LongMemEval. A natural reviewer worry is that the dual probe's Δ might reduce to a known long-context-recall failure. The external probe rules this reading out for the same model at the same order of context length: the pre-registered three-family external probe (Section 4.5) found every recovered cell scoring perfectly, including the DeepSeek-v4-Flash cells whose substrate $|\Delta|$ was largest on the original pilot. The substrate dissociation is therefore distinguishable from the generic retrieval-under-load failure that single-axis memory benchmarks measure; a deployment monitor that only watched verbal-recall health would have judged the dissociated cells fine.

Recommendation, with caveats. The dual probe is implementable on any structured-output agent loop that emits both prose and a separately addressable action surface. The substrate-side machinery here is one instantiation, and its determinism makes the probe reproducible to a SHA-256 hash in the no-LLM control, but the probe itself does not depend on the substrate. The recommendation we draw from the pilot is that a paired action-grounded probe is worth co-monitoring with verbal-only memory health checks on the kind of structured-output agent loop we tested; whether the recommendation generalises to production deployments outside the colony domain is the open question Section 6 flags as the gating follow-on.

6 Limitations

The eighteen-cell pilot is narrower than the matrix needed to characterise the dual probe definitively: three seeds per scenario at two horizons on a single model family cannot support strong claims about the scenario-stratified short-horizon ordering, the horizon-attenuation pattern, or the independence of dissociation magnitude from external verbal recall. We report the pilot as a threshold-crossing existence and scenario-stratification result, and the H2 independence finding as *supported on the pre-registered three-family probe* rather than as a general property of the dual probe. The pre-registered wider design (three model families, three scenarios, five seeds, four horizons) is the subset we could run under the per-model API availability we had during the study window; the cross-family extension and the full pre-registered matrix are the immediate experimental priorities for follow-on work.

A specific operational limitation: the external verbal-recall probe recovered sixteen of eighteen cells with real-LLM responses; two Claude-Sonnet-4.6 60-minute cells failed when the proxy wallet was exhausted by the $\sim$25,000-token prompts. The methodological circularity of the earlier AFR-as-its-own-proxy formulation is resolved by the external probe, but the perfect- recall ceiling effect on the recovered cells is itself a soft limit: a harder probe (more anchors, harder distractors, longer context) might puncture the ceiling and produce a measurable non-zero correlation. We frame the H2 outcome as independence supported *by the absence of observable coupling at the LongMemEval-style probe's difficulty level*, and identify a harder external probe as the next experimental step.

A more fundamental concern is that the dual probe operates inside a substrate we designed to host it. The independent resource dynamics (Section 3.1) rule out the most degenerate failure mode—AGR collapsing into a JSON-token echo—but do not address whether the dissociation generalises beyond the colony-simulation domain. A stronger validation would re-instantiate the dual probe in a substrate we did not author—a dialogue agent loop, a robotics planning loop, or an existing structured-output agent benchmark instrumented post-hoc—and confirm that the same probe definitions surface comparable dissociation patterns. We expect generalisation to follow from the structural property the substrate provides (a verbal channel and an action channel sharing a single model state across cadences) rather than the specific colony content, but the test is open. We identify cross-substrate validation as the second priority for follow-on work, alongside the broader matrix sweep; the present findings should be read as conditional on it.

7 Conclusion

We contribute a reproducible assay—the dual probe plus a deterministic substrate that wires the action channel into observable worker behaviour—and a substrate-coupled pilot of DeepSeek-v4-Flash across three scenarios in which dissociation crosses the pre-registered threshold in two of eighteen cells, separates by scenario at the short horizon, and attenuates roughly $2.5\times$ at the long horizon. A pre-registered external LongMemEval-style probe on three model families at matched context lengths recalls every recovered anchor perfectly, ruling out the simplest long-context-recall reading at the probe's difficulty level. We do not yet establish a general agent failure mode: the cross-family extension and the cross-substrate validation are the gating open questions.

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A Reproducibility

The substrate and the dual probe are released as an open-source Python package targeting Python 3.11 and above. Determinism is verified on three tiers. Tier 1 fixes the no-LLM control: identical seeds reproduce identical world states bit-for-bit across runs, verified by SHA-256 hashes of the world-state stream and confirmed across two independent runs on the same operating system. Tier 2 fixes a record-replay cache adapter that replays prior LLM responses byte-for-byte, isolating the substrate's contribution to the trajectory from the LLM's sampling. Tier 3 reports the expectation under live-LLM stationarity: with a fixed model snapshot and deterministic decoding, per-cell statistics remain within a documented tolerance band across re-runs.

Quick start. The package exposes a Typer-based command-line entry point. A single pilot cell is reproduced by

```
project-utopia run\
-experiment E1 -scenarios s_plains\
-seeds 0xBEEF,0xC0FFEE,0xCAFE\
-duration-sec 3600 -cadence-multiplier
6.0\
-out output/E1.ndjson
```

The NDJSON output stream contains per-cell AFR, AGR, Δ, and drift values along with the raw LLM responses cached for replay. Model snapshot identifiers, prompt schemas, and the anchor list used in the pilot are pinned in the released configuration files.

Software environment. Python $\geq$ 3.11 is required. All randomness flows through a single PCG64 generator with keyed BLAKE2b sub-seed derivation; the release does not depend on the legacy `random` or `numpy.random` global generators. The package ships with an automated determinism audit (`project-utopia-determinism`) that re-runs the three tiers and compares hashes against the pinned reference values, and an RNG coverage audit (`project-utopia-rng-audit`) that asserts every stochastic call site is reached by the seeded generator.

B Anchor Injection Protocol

The five anchors injected at $t = 0$ in the pilot are summarised in Table 3. Each anchor pairs a single-keyword verbal token—chosen short and concrete to maximise the chance of verbatim re-emission—with an implicit goal whose action key is addressable on the substrate's action surface.

Verbal token	Implicit-goal action key
warehouse	deliver
food	farm
wood	wood
guard	guard_engage
quarry	quarry

Table 3: Anchor pairs injected at $t = 0$ for the pilot.

The on-disk anchor representation is JSON of the form

```
{ "verbal_token": "warehouse",
   "implicit_goal": {
     "action": "deliver",
     "target": "warehouse"
   },
   "anchor_pos": [12, 8] }
```

written into the LLM's persistent world-summary memory at $t = 0$ without further instruction. The substrate does not surface the anchor pair to the model as a directive; the model must autonomously integrate the pair through its own summarisation and policy-weight behaviour.

Calibration of $|A| = 5$. The anchor count was calibrated during substrate development. With $|A| = 3$, AFR saturated at 1.0 for every model family at the short horizon, eliminating discriminative power in the verbal channel. With $|A| = 7$, the structured-output validity rate of the smallest open-weight model dropped below 80%, contaminating both probes with parsing-driven zeros. $|A| = 5$ preserved discrimination in the verbal probe while maintaining structured output integrity across all three families we tested.

C Prompt Templates

Each of the four channels described in Section 3.1 is driven by a dedicated prompt template. The environment-director prompt requests weather, event, and faction-tension updates as a JSON object validated against a pydantic schema. The npc-policy prompt requests per-group intent weights and target-priority lists on the clamped $[0, 3]$ scale. The strategic-plan prompt requests a long-form primary goal, a list of constraints, and a free-form summary intended as the verbal channel of the agent. The colony-agent prompt requests a per-step build plan following the Perceive–Plan–Ground–Execute–Evaluate–Reflect pattern, with each phase emitted as a separately validated JSON sub-object.

The four prompts share a common observation envelope that summarises recent world state, scenario blueprints, and the persistent world-summary memory into which anchors have been injected. Outputs are validated and clamped through a shared guardrail layer before they act on the world. Verbatim prompt templates and the pydantic response-schema definitions are released with the code package; we omit them from the paper body to preserve length, and refer the reader to the released repository for the exact strings used in the pilot.